

MRPL Refinery — Standard Operating Procedure: Main Fractionator Column V-220

Document Control

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Equipment Specification

Parameter	Value
Equipment Tag	V-220
Equipment Type	Atmospheric Main Fractionator, 42 Trays
Service	Crude Oil Atmospheric Distillation
Design Pressure	4.5 bar g
Design Overhead Temperature	125 C

Normal Operating Limits

Parameter	Normal Range	High Alarm	High-High / Trip
Overhead Pressure	3.2-3.8 bar g	4.0 bar g	4.2 bar g
Overhead Temperature	110-118 C	122 C	125 C
Bottoms Level	45-60 percent	70 percent	80 percent
Reflux Ratio	1.8-2.2	n/a	n/a

Flooding Indication and Response

A sustained rise in column differential pressure (DP) above 0.35 bar across the tray section, combined with erratic overhead temperature and reduced product draw-off rates, is indicative of tray flooding. Upon flooding indication: (1) Reduce feed rate by 10-15 percent. (2) Reduce reflux ratio incrementally while monitoring overhead composition. (3) If DP does not stabilize within 30 minutes, escalate to Unit Engineer for further feed reduction or planned rate cutback. Do not attempt to correct flooding solely by increasing reboiler duty, as this can exacerbate vapor loading.

Reference

This SOP should be read in conjunction with P&ID; CDU-001 (Main Fractionator) and the Process Design Basis for V-220. Any deviation from these limits during abnormal crude assay changes must be reviewed with Process Engineering before adjusting control setpoints.